1. Is there any required format for submission of the response?
    a. See Section L on Page 126 for proposal formatting.

2. What content should be presented?
    a. Section L page 126 to 131 and Section M page 134 to 141 .

3. Do you require a budgetary estimate of costs currently?
    a. All information on Cost/Price factor can be found on pages 140 to 141 (Section M) and pages 130 to 131 (Section L)

4. What is the method of submission? Should this be via email?
    a. Submission via email. Section L page 127

5. Sensor Requirements – REQ-SR-17, 18, and 19 are somewhat unclear as to exactly what they mean. When precision based on how far the vehicle moves is mentioned, what is this relative to? Is this based on the course reference point established in REQ-SR-16? If so, why is it relevant how far the vehicle has moved, when the precision/accuracy is being determined based on a specific reference point? Please advise and clarify.
    a. The current profilometer functions with an Inertial Navigation system to calculate position as well as compensate for the vertical movement of the vehicle as the suspension moves offroad. As the vehicle moves, errors build up in the profile point cloud and the relative precision between two points decreases. For example, in a scan of a 1 mile straight long test course, two points collected within 10 feet of each other will have less error than points collected 1000 feet from each other.

6. Scan Width (REQ-SR-12) – The scan width must be sufficient to cover the width of endurance and performance courses at ATC.
    a. Threshold (minimum level of performance that is considered achievable at low-to-moderate risk) = The scan width will be 12 feet or greater
    b. O (Level of performance which is the ultimate desired operational goal) = The scan width will be variable between 12 and 20 feet

    How important is it to meet the objective, versus just the threshold? What operational goals does this achieve?

    The full 20 ft. is very long, providing stability to the platform over this length is very difficult.

    a. Some of our test courses are considerably wider than 12 feet and it is important to us to be able to classify all the features on the course as test vehicles do not travel on the exact same tracks as the profilometer does.
        i. ACC Note: Any requirement ratings are presented within the evaluation criteria and supporting documentation.

7. Design Analysis – How critical is it for you to view the terrain after the vehicle goes over the full 20 ft? Is it acceptable to just look at the two areas where the wheels went over the terrain?
    a. It is important to have a comprehensive analysis of the entire course to make maintenance and testing decisions. As per REQ-DR-07 we are looking for software that can detect and record terrain features such as potholes over the full course scan. Only using data from the wheel paths will miss key portions of the terrain.

8. Operating Conditions – Is the system to operate just at day? Or at night also? Should the system be expected to operate in all weather conditions (rain, fog, snow, dry, etc)? Please specify these more clearly.
    a. The system is intended to operate during the day and is expected to function in all weather conditions. A portion of the environmental conditions that we will operate under is dust and heavy mud. We do not normally operate during heavy rain or snow and understand that data may not be within specifications during those times.

9. There is documentation about the current system that is being used. What about the current system works and what does not work? (i.e. what are the strengths and weaknesses of the present systems)?
    a. The biggest limitation on the current system is that while the full point cloud is collected, the analysis is mainly done on the two tracks behind the wheels of the vehicle. This leaves a large portion of the course conditions out of the results that are sent out to our testing management. The other main issue is reliability; the system does not have the robust construction to withstand years of vibrations and dirt from harsh offroad driving.